Efficient Sentiment Classification using GPT-2, LoRA, and Knowledge Distillation

Mohamed Islam[1], Youssef Al Hussaini[2], and Ahmed Haytham[3]

Computer Science and Engineering, Alamein International University

[1]mohamed.elshourbagy.2023@aiu.edu.eg

[2]youssef.alhussaini.2024@aiu.edu.eg

[3]ahmed.hanafi.2024@aiu.edu.eg

Original CS224N Reference: "Sentiment Classification using GPT-2 Representations," Stanford CS224N Default Project by Sahaj Saini.

Abstract

Large pre-trained language models have established state-of-the-art performance across Natural Language Processing (NLP) tasks, but their full fine-tuning requires prohibitive computational resources. This work proposes a parameter-efficient pipeline for sentiment classification that optimizes a GPT-2 baseline without relying on massive computational overhead. The approach integrates Low-Rank Adaptation (LoRA) with teacher-student knowledge distillation, using a fully fine-tuned GPT-2 model as the teacher and a DistilGPT-2 model with LoRA adapters as the student. Standard final-token extraction is replaced with a hybrid pooling strategy, combining 80% attention pooling with 20% mean pooling, to better capture sentence context. The pipeline is evaluated on the Stanford Sentiment Treebank (SST-5) and the IMDB dataset. Results show that the LoRA-distilled student retains highly competitive performance compared to the fully fine-tuned teacher while training only 2.15% of the total parameters. This demonstrates that combining parameter-efficient fine-tuning with knowledge distillation offers a viable, low-cost alternative for deploying robust transformer-based sentiment classifiers.

Keywords: Sentiment Classification; Low-Rank Adaptation (LoRA); Knowledge Distillation; GPT-2; DistilGPT-2

Introduction

Transformer-based architectures have revolutionized Natural Language Processing, setting new benchmarks across various tasks, including sentiment analysis. However, as model sizes scale, full fine-tuning becomes increasingly impractical due to immense memory and computational constraints. The baseline approach for CS224N sentiment classification projects typically involves full parameter updates on models like GPT-2, which is highly inefficient for rapid iteration and deployment on edge or resource-constrained environments.

The primary objective of this project is not to surpass state-of-the-art accuracy, but rather to maximize parameter efficiency, reduce computational costs, and improve training stability while maintaining competitive performance. To this end, a parameter-efficient fine-tuning (PEFT) methodology utilizing Low-Rank Adaptation (LoRA) is introduced. Instead of training all weights, LoRA freezes the pre-trained model and injects trainable rank-decomposition matrices into the transformer layers.

To prevent the performance degradation typically associated with smaller or heavily constrained models, teacher-student knowledge distillation is incorporated, where a fully fine-tuned GPT-2 teacher guides the LoRA-adapted DistilGPT-2 student. Finally, to optimize the textual representation extracted from the decoder-only architecture, a hybrid pooling mechanism is introduced. By addressing the bottlenecks of full fine-tuning, this report presents a scalable, efficient, and robust solution for transformer-based sentiment classification.

Transformer models rely on self-attention mechanisms to capture long-range dependencies in text. Previous CS224N baseline projects have successfully adapted auto-regressive models like GPT-2 for classification by appending a linear head to the final sequence token; however, this full fine-tuning approach modifies millions of parameters, leading to high memory overhead (Radford et al. 2019). To mitigate the costs of full fine-tuning, LoRA was introduced by Hu et al. (2021), who hypothesized that the change in weights during model adaptation has a low intrinsic rank; by optimizing dense layers indirectly through low-rank matrices, LoRA significantly reduces the number of trainable parameters while achieving parity with full fine-tuning on many NLP benchmarks. Knowledge distillation, first formalized by Hinton et al. (2015), transfers knowledge from a large, cumbersome model (teacher) to a smaller, more efficient model (student) by training the student to mimic the teacher's soft probability distributions; in NLP this has been highly successful for compressing models such as BERT into DistilBERT (Sanh et al. 2019). This work combines distillation with LoRA to maximize both parameter and inference efficiency.

Materials and methods

Baseline configuration

The original baseline utilizes a standard GPT-2 sentiment classifier. Two baseline configurations are evaluated: a Frozen Baseline, in which the entire GPT-2 feature extractor is frozen and only a standard classification head is trained; and a Full Fine-Tuning Baseline, in which all parameters of the GPT-2 model are updated using the AdamW optimizer. Both baselines rely on the final-token representation to predict sentiment.

Proposed architecture

The proposed approach drastically reduces the trainable parameter count and consists of the following pipeline:

- Tokenization: input text is processed with minimal preprocessing, using dynamic padding and attention masks to preserve raw contextual and syntactic information.
- Encoder: DistilGPT-2 is used as the base student architecture.
- LoRA adapters: the base model is frozen, and LoRA adapters are injected into the query and value projection matrices, using rank $r = 16$, alpha $\alpha = 32$, and a dropout rate of 0.1.
- Hybrid pooling: rather than relying solely on the last token, a hybrid pooling strategy combining attention pooling (80%) and mean pooling (20%) is used to create a richer sentence-level embedding.
- Classification head: the pooled representation passes through a LayerNorm and Dropout layer before the final sentiment prediction.

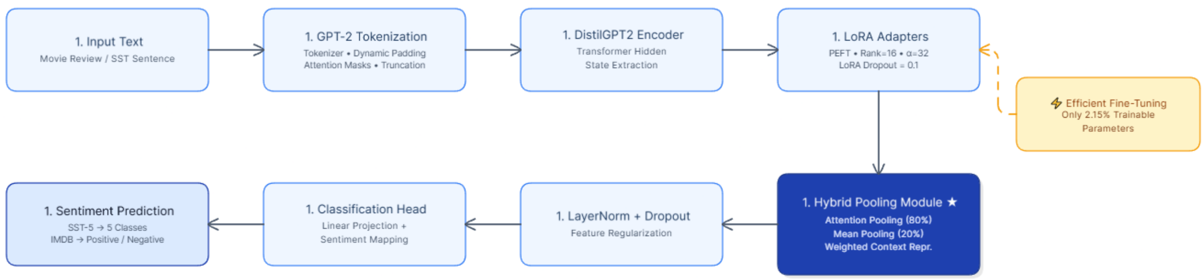

Figure 1. The proposed pipeline illustrating the flow from tokenization to the hybrid pooling mechanism and the LoRA-adapted student model.

Distillation and loss functions

The student model does not directly copy the teacher's weights. Knowledge is transferred using soft probability distributions via KL-divergence. The total distillation loss L is a weighted sum of the supervised hard classification loss (L_hard) and the teacher-student distillation loss (L_KL): L = α·L_hard + (1 − α)·L_KL. The distillation temperature is set to 2.0 and α to 0.7. For L_hard, standard cross entropy with label smoothing is applied for the binary IMDB dataset; for the highly imbalanced SST-5 dataset, focal loss is used to prevent the model from being overwhelmed by majority classes.

Optimization techniques

To improve training stability and efficiency, the AdamW optimizer is paired with a cosine learning-rate scheduler with warmup. Mixed-precision training (AMP), gradient accumulation, gradient checkpointing, and early stopping are also employed. Weighted random sampling is used during data loading to counteract class imbalance.

Datasets

Two datasets are used for evaluation:

- SST-5 (Stanford Sentiment Treebank): a fine-grained sentiment classification dataset with five classes (very negative, negative, neutral, positive, very positive). This dataset is highly ambiguous and difficult due to subtle sentiment distinctions.
- IMDB: a large-scale movie review dataset for binary sentiment classification (positive, negative), featuring longer texts and stronger sentiment cues, making it a comparatively easier classification task.

Experimental setup

Three primary configurations are compared: the Frozen Baseline, the Full Fine-Tuning Teacher (GPT-2), and the proposed LoRA Distilled Student (DistilGPT-2 + LoRA). Models are evaluated using accuracy and macro F1 scores.

Results and discussion

Table 1 and Table 2 summarize the final test results on the SST-5 and IMDB datasets, respectively.

Table 1. Final test results on the SST-5 dataset.

Model	Accuracy	Macro F1
Frozen Baseline	0.3294	0.2693
Full GPT-2 Teacher	0.4950	0.4938
LoRA Distilled Student	0.4290	0.4232

Table 2. Final test results on the IMDB dataset.

Model	Accuracy	Macro F1
Frozen Baseline	0.7958	0.7955
Full GPT-2 Teacher	0.9245	0.9245
LoRA Distilled Student	0.8962	0.8962

Parameter efficiency

The core success of the proposed method lies in parameter efficiency. The Full GPT-2 Teacher requires updating 100% of its parameters, whereas the LoRA Distilled Student model utilizes only 2.15% trainable parameters. Despite this large reduction in computational overhead, the student model retains the vast majority of the teacher's performance (e.g., an 0.8962 macro F1 compared with the teacher's 0.9245 on IMDB). This indicates that LoRA is highly effective at reducing memory usage and computational cost without sacrificing robustness.

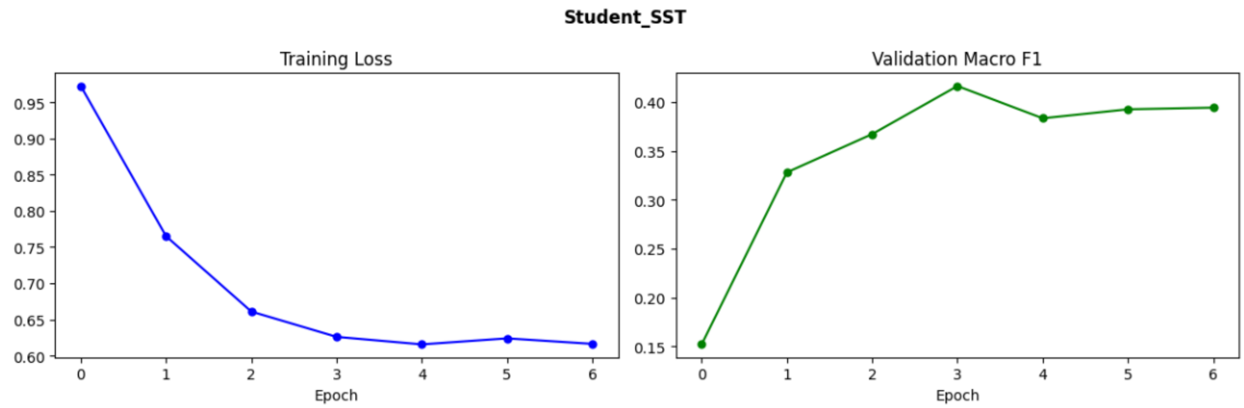

Figure 2. Training and validation curves for the LoRA Distilled Student model.

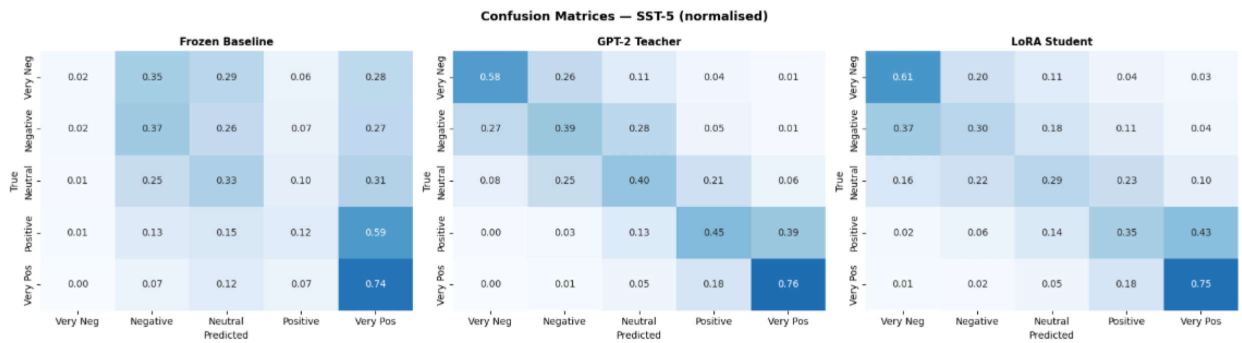

Figure 3. Confusion matrices comparing the Baseline, Teacher, and Student on SST-5.

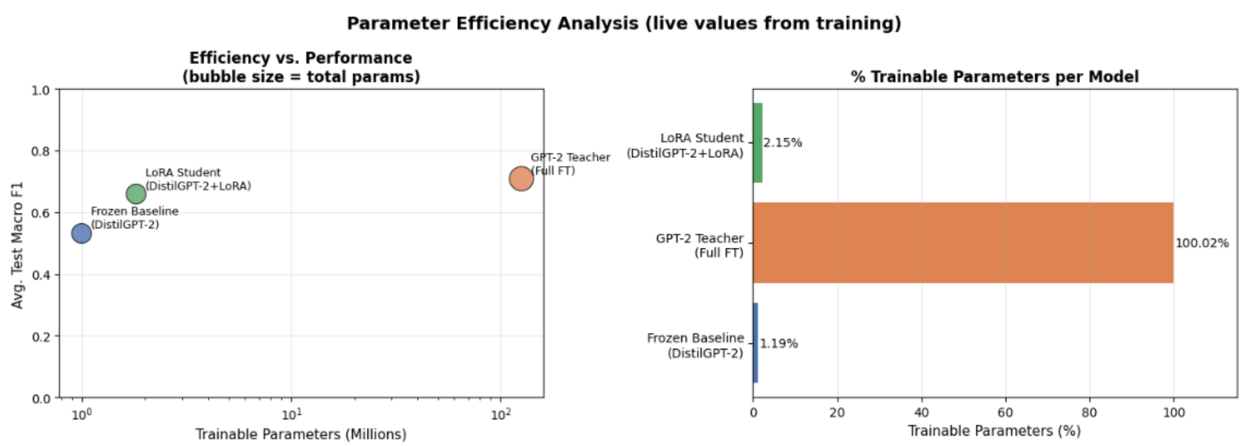

Figure 4. Tradeoff between trainable parameters and macro F1 score across the three configurations.

Dataset complexity: SST-5 versus IMDB

The performance disparity between the two datasets highlights the inherent challenges of fine-grained classification. SST-5 proved significantly harder due to its five-class structure and heavy reliance on semantic ambiguity and subtle sentiment distinctions (e.g., differentiating between “neutral” and “slightly positive”). By contrast, the models performed exceptionally well on IMDB: its binary nature, combined with stronger, more explicit sentiment cues and a larger volume of data, allowed for clearer sentiment separation.

Ablation and architectural contributions

The Frozen Baseline served as a reproduction baseline and yielded poor results due to the inability of the pretrained weights to adapt to the specific sentiment spaces, while the Full GPT-2 Teacher established a strong performance ceiling. Knowledge distillation effectively transferred the teacher's generalized feature-extraction capabilities to the student, preventing it from overfitting on the hard labels. The hybrid pooling approach proved crucial, since auto-regressive models typically rely entirely on the final token, which can dilute information from earlier in the sentence; combining 80% attention pooling with 20% mean pooling ensured that the classifier received a dense, contextually rich embedding. The decision to rely on minimal preprocessing was also validated, since removing stopwords or punctuation strips away syntactic context that transformer attention mechanisms rely on to understand sentiment.

Limitations

While successful in improving efficiency, the approach has limitations. The inherent difficulty of the SST-5 dataset constrained absolute accuracy across all models, and computational constraints limited the scope of the hyperparameter search. Future work could explore dynamic LoRA rank allocation or extend the distillation process to intermediate hidden layers rather than only the final logits.

Conclusion

This report presented an efficient transformer fine-tuning pipeline for sentiment analysis. By integrating Low-Rank Adaptation (LoRA) with teacher-student knowledge distillation, a student model was developed that achieves competitive performance while drastically reducing computational overhead, training only 2.15% of the total parameters. The hybrid pooling mechanism and optimization strategies further stabilized training. Overall, this approach demonstrates that state-of-the-art architectures can be adapted for complex NLP tasks efficiently, making them viable for real-world, resource-constrained applications.

Data availability

This study relies on two publicly available datasets: the Stanford Sentiment Treebank (SST-5) and the IMDB movie review dataset, both of which are standard, publicly accessible NLP benchmarks.

Acknowledgments

Not applicable.

Funding

This project was completed as coursework and received no external funding.

Conflicts of interest

The authors declare no conflicts of interest.

Author contributions

- Mohamed Islam: model implementation, experimentation with LoRA adapters, optimization loop debugging.
- Youssef Al Hussaini: data preprocessing, implementation of hybrid pooling, generation of training visualizations.
- Ahmed Haytham: knowledge distillation implementation, error analysis, comprehensive report writing and formatting.

Literature cited

Hinton G, Vinyals O, Dean J. 2015. Distilling the knowledge in a neural network. arXiv preprint arXiv:1503.02531.

Hu EJ, Shen Y, Wallis P, Allen-Zhu Z, Li Y, Wang S, Wang L, Chen W. 2021. LoRA: low-rank adaptation of large language models. arXiv preprint arXiv:2106.09685.

Radford A, Wu J, Child R, Luan D, Amodei D, Sutskever I. 2019. Language models are unsupervised multitask learners. OpenAI blog. 1(8):9.

Sanh V, Debut L, Chaumond J, Wolf T. 2019. DistilBERT, a distilled version of BERT: smaller, faster, cheaper and lighter. arXiv preprint arXiv:1910.01108.